

17-1a A Duopoly Example

To understand the behavior of oligopolies, let’s consider an oligopoly with only two members, called a *duopoly*. Duopoly is the simplest type of oligopoly. Oligopolies with three or more members face the same problems as duopolies, so we do not lose much by starting with the simpler case.

Imagine a town in which only two residents, Jack and Jill, own wells that produce water safe for drinking. Each Saturday, Jack and Jill decide how many gallons of water to pump, bring the water to town, and sell it for whatever price the market will bear. To keep things simple, suppose that Jack and Jill can pump as much water as they want without cost. That is, the marginal cost of water equals zero.

Table 1 shows the town’s demand schedule for water. The first column shows the total quantity demanded, and the second column shows the price. If the two well owners sell a total of **10** gallons of water, water goes for **\$110** a gallon. If they sell a total of **20** gallons, the price falls to **\$100** a gallon. And so on. If you graphed these two columns of numbers, you would get a standard downward-sloping demand curve.

Table 1		
The Demand Schedule for Water		
Quantity	Price	Total Revenue (and total profit)
0 gallons	\$120	\$ 0
10	110	1,100
20	100	2,000
30	90	2,700
40	80	3,200
50	70	3,500

Quantity	Price	Total Revenue (and total profit)
60	60	3,600
70	50	3,500
80	40	3,200
90	30	2,700
100	20	2,000
110	10	1,100
120	0	0

The last column in [Table 1](#) shows total revenue from the sale of water. It equals the quantity sold times the price. Because there is no cost to pumping water, the total revenue of the two producers equals their total profit.

Let's now consider how the organization of the town's water industry affects the price of water and the quantity sold.